

UrbanPulse: Satellite Urban Sprawl & Land Cover Intelligence

High-resolution multi-temporal satellite analytics engine tracking urban sprawl, concentric ring densification, and Shannon entropy across metropolitan regions using Sentinel-2 and ESA WorldCover.

[Live Web Dashboard](#)[GitHub Repository](#)[Generated: 2026-10-01 14:08 UTC](#)[Git Commit:4bb9764e](#)[✓ CI/CD: Passing](#)

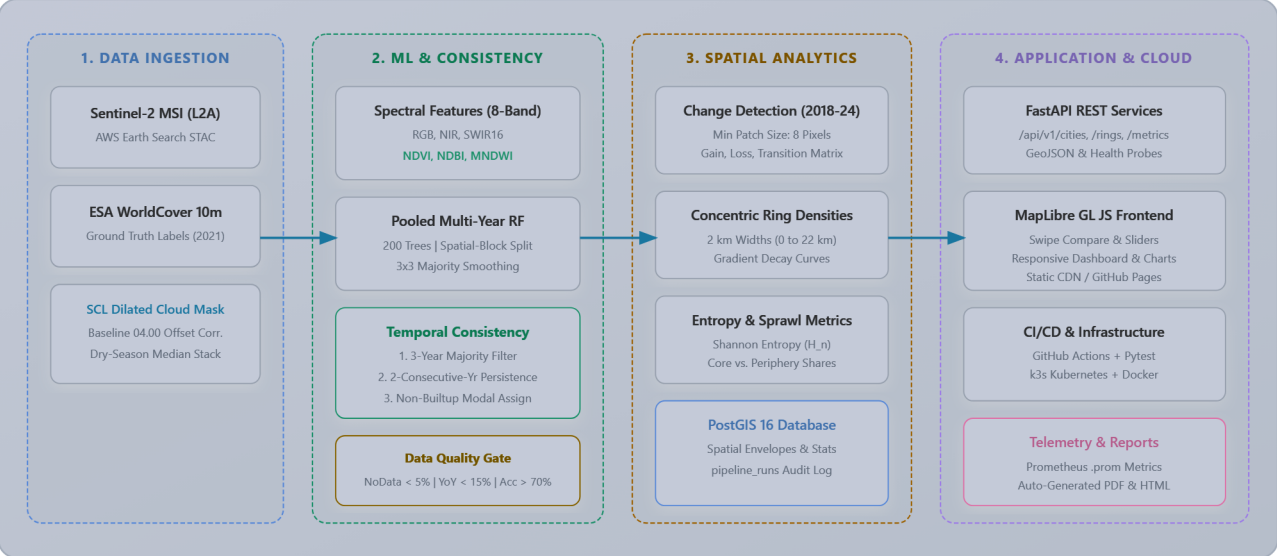
1. Problem & Approach

Rapid urbanization across major metropolitan regions frequently outpaces civic planning frameworks, leading to unchecked peripheral sprawl, fragmentation of agricultural zones, and loss of critical blue-green infrastructure. Monitoring these dynamics using ad-hoc, single-scene satellite imagery often introduces severe seasonal phenological confusion, cloud contamination, and sensor baseline artifacts that corrupt time-series measurements.

UrbanPulse was engineered as an automated, reproducible Earth Observation (EO) pipeline. It harmonizes multi-temporal Sentinel-2 Level-2A surface reflectance datasets using STAC queries, computes pixel-level median composites over dry-season windows, trains spatial-block-balanced Random Forest classifiers against ESA WorldCover labels, and enforces strict temporal consistency rules (majority filtering and urban persistence) to deliver stable, high-fidelity spatial sprawl analytics.

✂ 2. System Architecture

The end-to-end data flow spanning satellite catalog ingestion, radiometric processing, machine learning classification, PostGIS spatial ingestion, and containerized deployment.





3. Data & Methodology

Earth Observation Data Specifications

- **Sensor:** Sentinel-2 Multi-Spectral Instrument (MSI), Level-2A Bottom-Of-Atmosphere (BOA) surface reflectance via AWS Earth Search STAC.
- **Spatial Resolution:** 60.0 m analytical pixel grid (EPSG:32643 UTM projection), with 10.0 m high-resolution visual previews.
- **Area of Interest (AOI):** 45.0 × 45.0 km bounding envelope centered on the municipal core (571,536 grid pixels per city).
- **Dry-Season Compositing:** November 1 to February 28 window (with automated 1-month seasonal widening if valid observations < 4).
- **Cloud & Shadow Masking:** SCL (Scene Classification Layer) masking cloud shadow, medium/high probability cloud, cirrus, and snow with 1-pixel binary dilation.
- **Radiometric Harmonization:** Explicit scale (0.0001) and offset (-0.1) normalization compensating for the European Space Agency Sentinel-2 Processing Baseline 04.00 shift post-January 2022.

Machine Learning & Temporal Consistency

- **Spectral Feature Stack (8 Bands):** Blue, Green, Red, NIR, SWIR16, plus spectral indices (NDVI, NDBI, MNDWI).
- **Supervised Classifier:** Random Forest (200 trees, `class_weight="balanced"`) pooled across reference years (2018, 2021, 2024).
- **Training Labels:** ESA WorldCover 10m v200 (2021) mapped to 5 standardized classes (Built-up, Vegetation, Water, Agriculture, Open Land).
- **Spatial Block Partitioning:** 8-fold non-overlapping spatial grid partition guaranteeing zero spatial autocorrelation between train and test folds.
- **Spatial Filtering:** 3 × 3 majority mode filter applied to raw raster predictions to eliminate salt-and-pepper noise.
- **Temporal Consistency Rules:**
 1. *Majority Rule:* Pixel is Built-up in year t only if Built-up in ≥ 2 of $(t-1, t, t+1)$.
 2. *Urban Persistence:* Once Built-up for 2 consecutive years, pixels permanently stay Built-up (urban land persistence assumption).
 3. *Modal Non-Built-Up:* Filtered false-urban pixels receive their multi-year non-built-up modal class.

4. Multi-City Analytical Results

Metropolitan Region: Ahmedabad

Model Evaluation & Accuracy (Pooled Multi-Year Random Forest)

OVERALL TEST ACCURACY

65.25%

Held-out Spatial Blocks

COHEN'S KAPPA (K)

0.5600

Inter-Class Agreement

PER-YEAR ACCURACIES

2018: 59.5% • 2021: 68.5% • 2024: 67.8%

Per-Class Precision, Recall & F1-Score

Class Name	Precision	Recall	F1-Score	Support
Built-up	72.2%	79.5%	75.6%	258
Vegetation	59.2%	52.2%	55.5%	270
Water	87.4%	89.0%	88.2%	219
Agriculture	52.3%	59.5%	55.6%	291
Open land	55.6%	42.6%	48.3%	162

5 × 5 Confusion Matrix

True \ Pred	Built	Veg	Water	Agri	Open
Built-	205	10	1	18	24
Vegeta	16	141	10	93	10
Water	6	8	195	9	1
Agricu	15	68	15	173	20
Open l	42	11	2	38	69

Temporal Cleanup Summary (Raw vs. Cleaned Built-up Footprint)

Year	Raw Built-up (km ²)	Clean Built-up (km ²)	Net Change (km ²)	Rule 1
2018	428.67 km ²	293.55 km ²	-135.12 km ²	
2019	320.38 km ²	368.85 km ²	+48.47 km ²	
2020	421.06 km ²	384.51 km ²	-36.55 km ²	
2021	413.61 km ²	414.07 km ²	+0.46 km ²	
2022	423.91 km ²	441.91 km ²	+18.00 km ²	
2023	504.42 km ²	474.54 km ²	-29.88 km ²	
2024	474.02 km ²	468.54 km ²	-5.48 km ²	

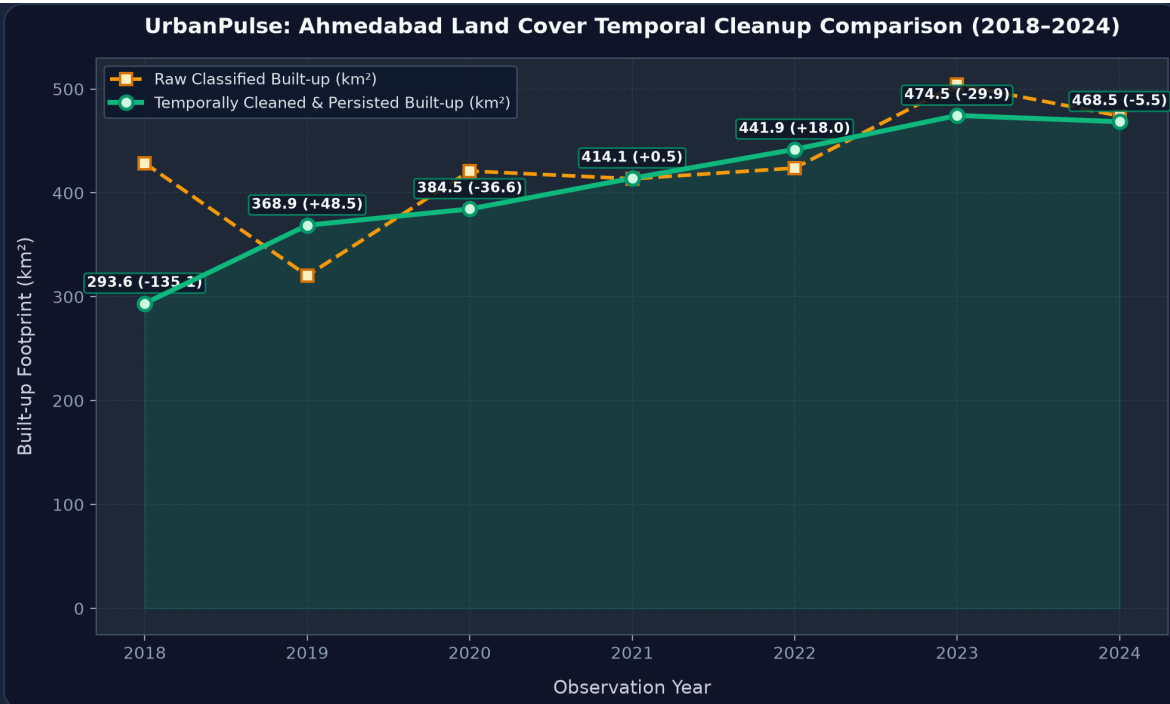


Figure: Ahmedabad Raw vs. Temporally Cleaned & Persisted Built-up Footprint (2018–2024).

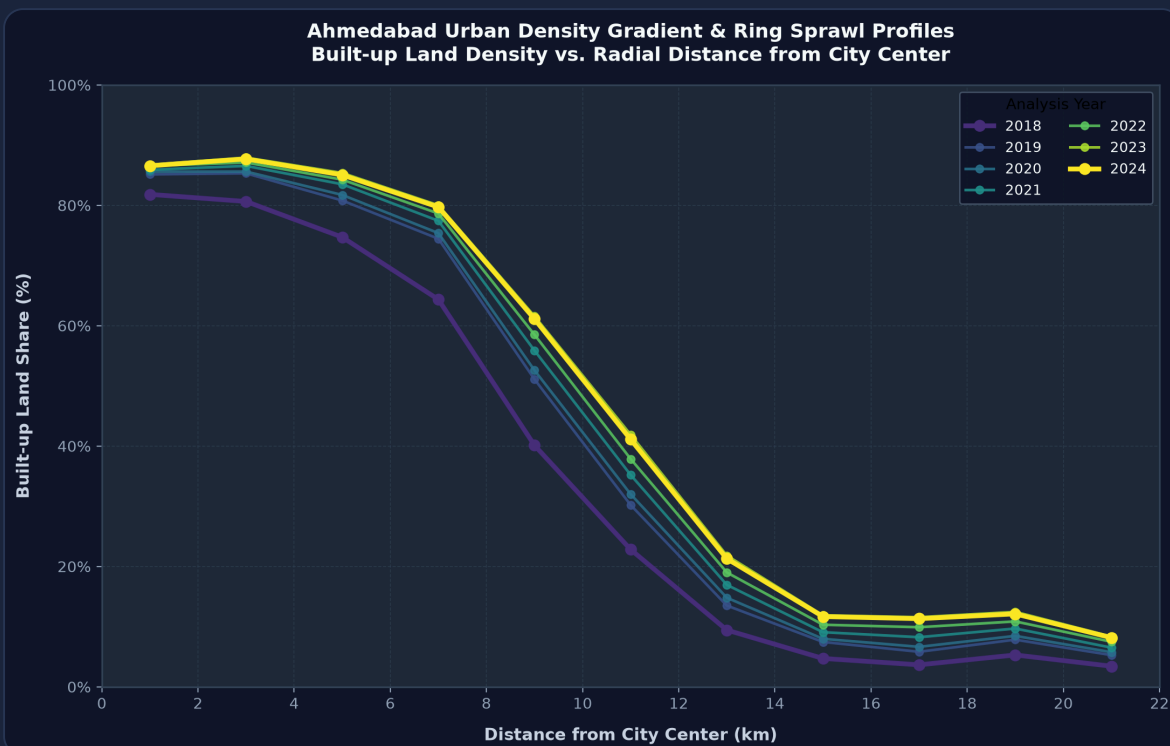


Figure: Ahmedabad Built-up Density vs Distance from CBD.

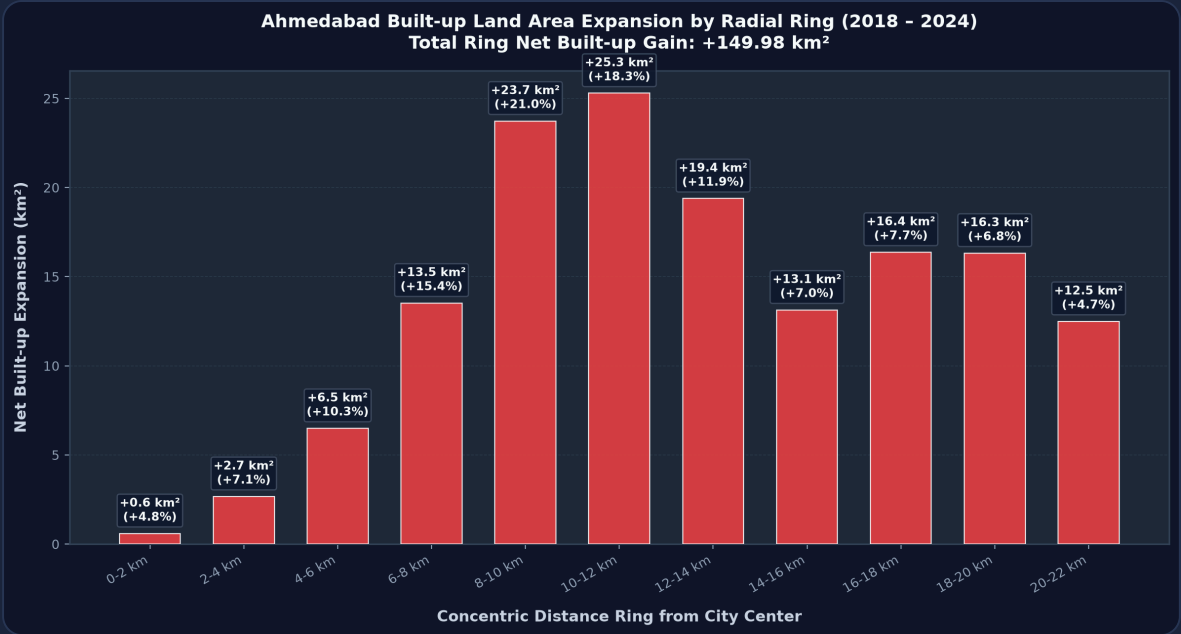
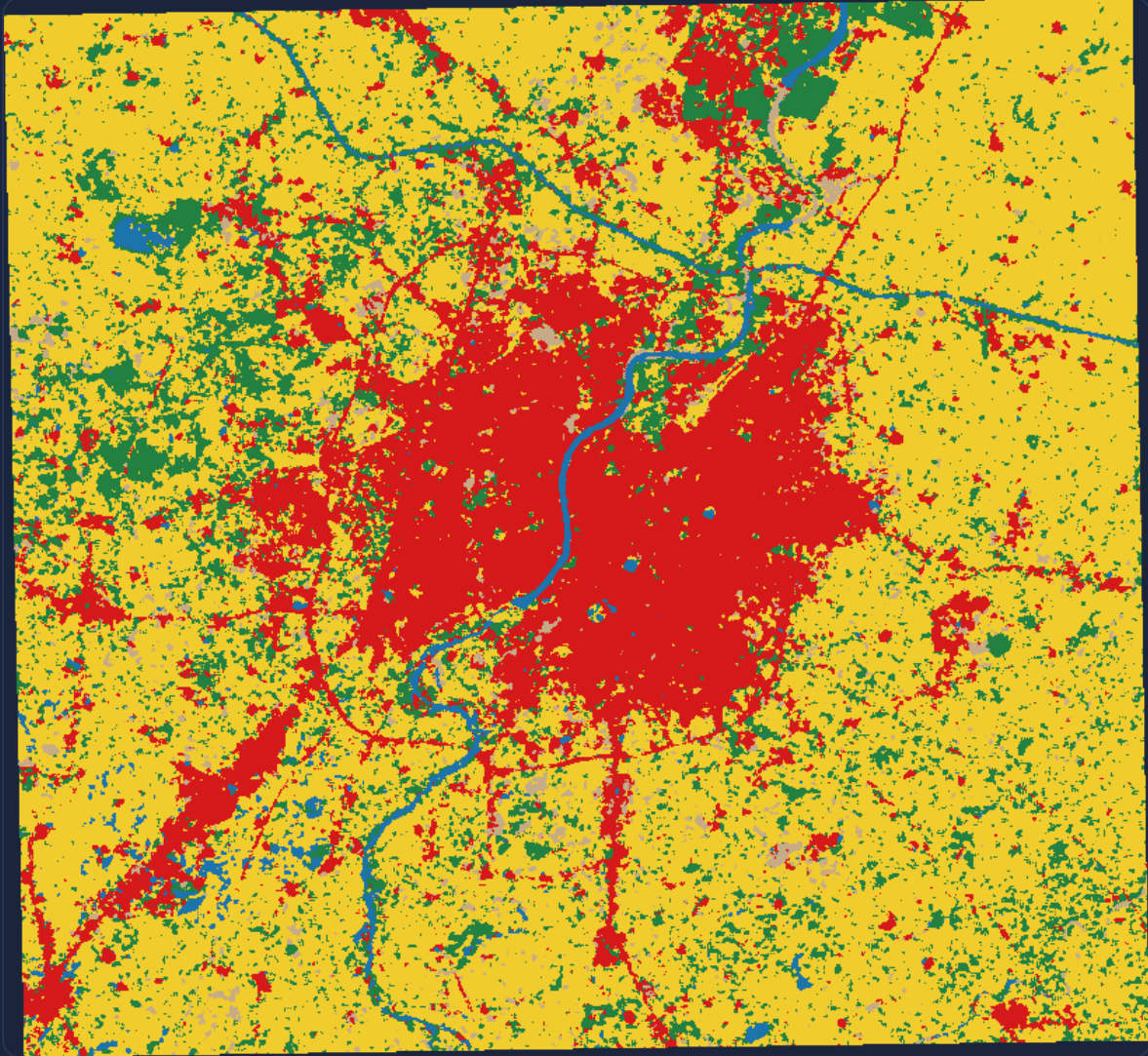


Figure: Ahmedabad Annual Ring Growth.

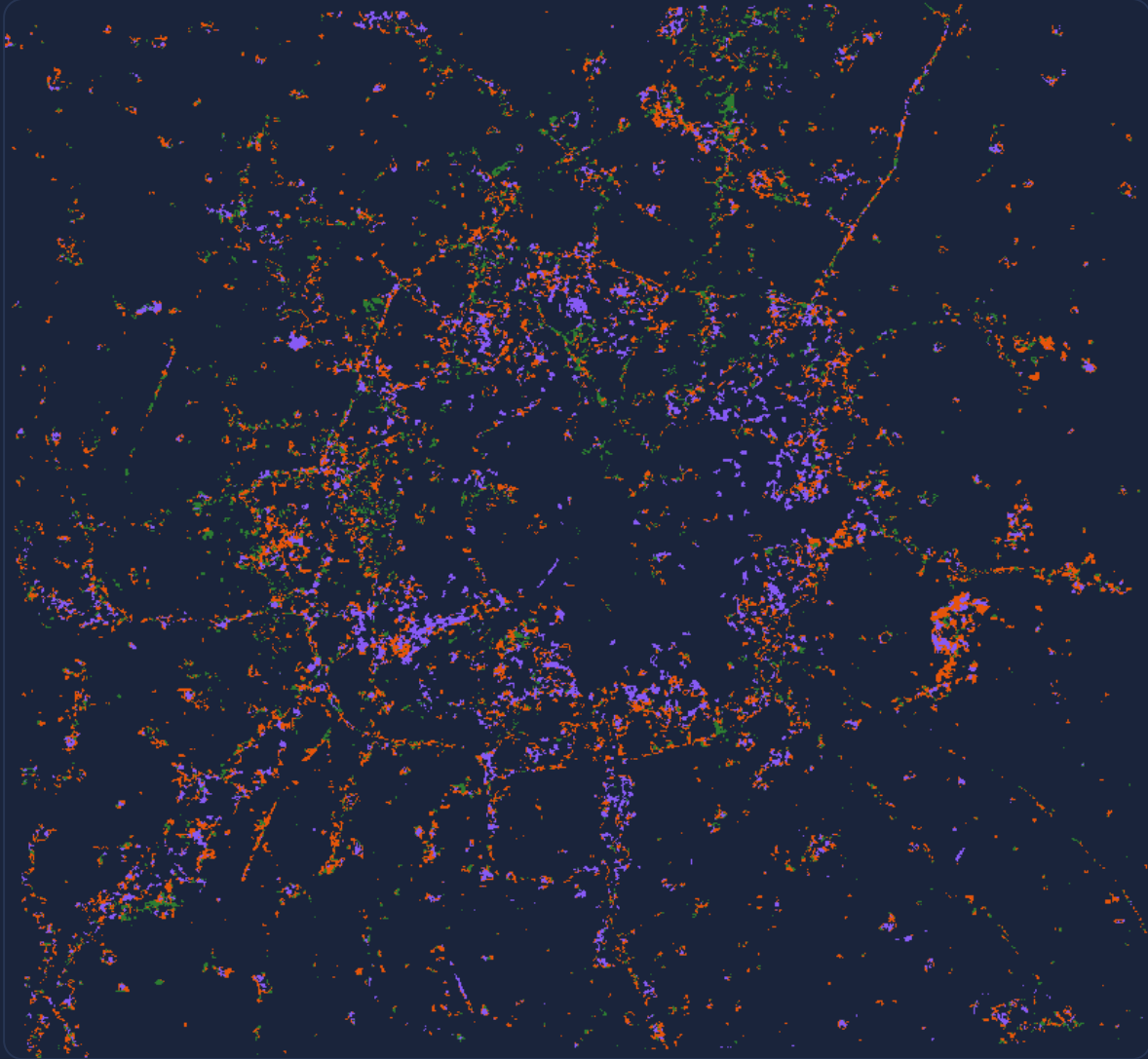
Spatial Sprawl, Entropy & Core-Periphery Distribution

Year	Built-up Area (km ²)	Annual Growth	CAGR vs '18	Shannon Entropy
2018	428.67 km ²	+0.00%	+0.00%	0
2019	320.38 km ²	-25.26%	-25.26%	0
2020	421.06 km ²	+31.42%	-0.89%	0
2021	413.61 km ²	-1.77%	-1.19%	0
2022	423.91 km ²	+2.49%	-0.28%	0
2023	504.42 km ²	+18.99%	+3.31%	0
2024	474.02 km ²	-6.03%	+1.69%	0

Spatial Output Maps



Classified Land Cover (2024)



Land Cover Change (2018 → 2024)

Metropolitan Region: Pune

Model Evaluation & Accuracy (Pooled Multi-Year Random Forest)

OVERALL TEST ACCURACY

81.96%

Held-out Spatial Blocks

COHEN'S KAPPA (K)

0.7582

Inter-Class Agreement

PER-YEAR ACCURACIES

2018: 73.4% • 2021: 86.0% • 2024: 86.5%

Per-Class Precision, Recall & F1-Score

Class Name	Precision	Recall	F1-Score	Support
Built-up	58.9%	59.3%	59.1%	123
Vegetation	84.1%	86.6%	85.3%	402
Water	100.0%	98.9%	99.5%	378
Agriculture	73.4%	78.3%	75.8%	240
Open land	47.3%	35.4%	40.5%	99

5 × 5 Confusion Matrix

True \ Pred	Built	Veg	Water	Agri	Open
Built-	73	8	0	19	23
Vegeta	6	348	0	35	13
Water	3	0	374	0	1
Agricu	14	36	0	188	2
Open l	28	22	0	14	35

Temporal Cleanup Summary (Raw vs. Cleaned Built-up Footprint)

Year	Raw Built-up (km ²)	Clean Built-up (km ²)	Net Change (km ²)	Rule 1
2018	410.11 km ²	248.77 km ²	-161.34 km ²	
2019	326.02 km ²	327.71 km ²	+1.69 km ²	
2020	332.13 km ²	356.19 km ²	+24.06 km ²	
2021	386.55 km ²	421.99 km ²	+35.44 km ²	
2022	692.99 km ²	443.75 km ²	-249.24 km ²	
2023	359.65 km ²	479.78 km ²	+120.13 km ²	
2024	406.65 km ²	448.31 km ²	+41.66 km ²	

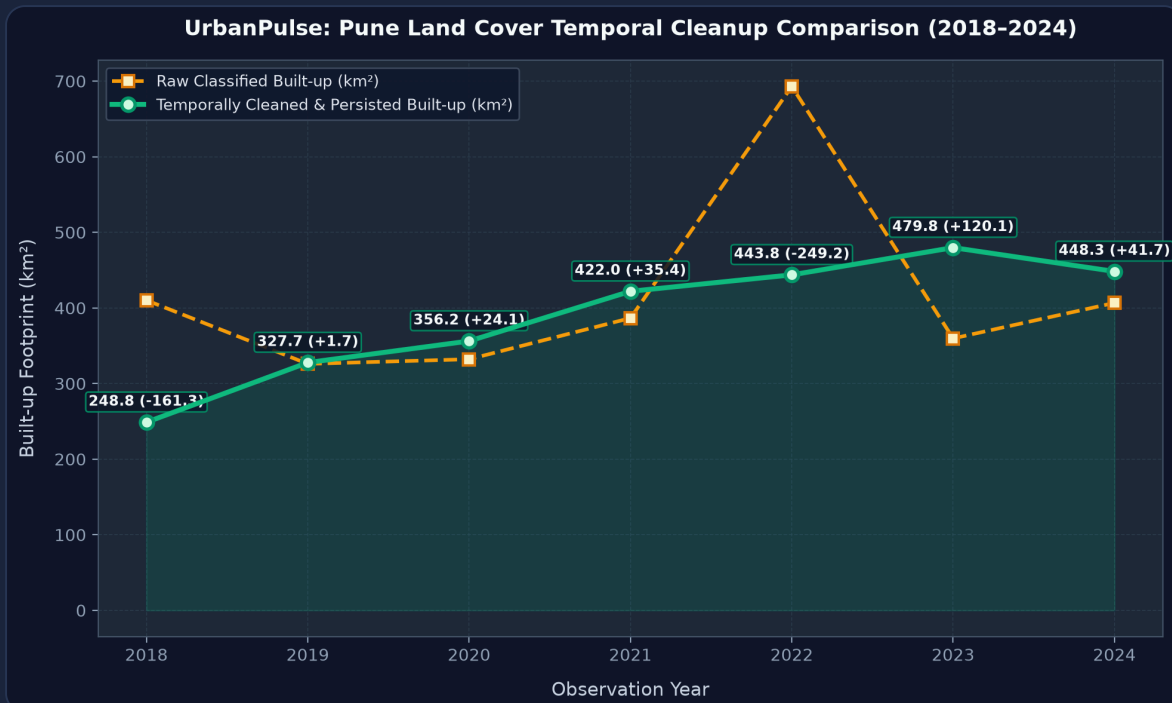


Figure: Pune Raw vs. Temporally Cleaned & Persisted Built-up Footprint (2018–2024).

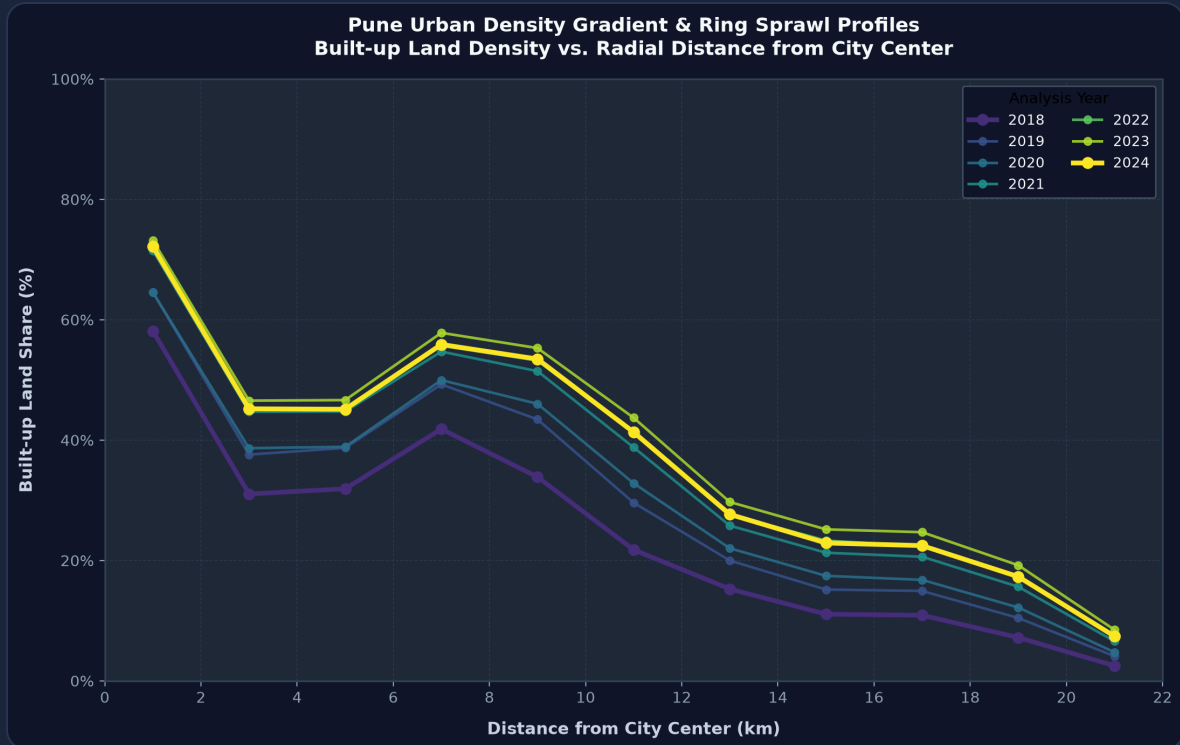


Figure: Pune Built-up Density vs Distance from CBD.

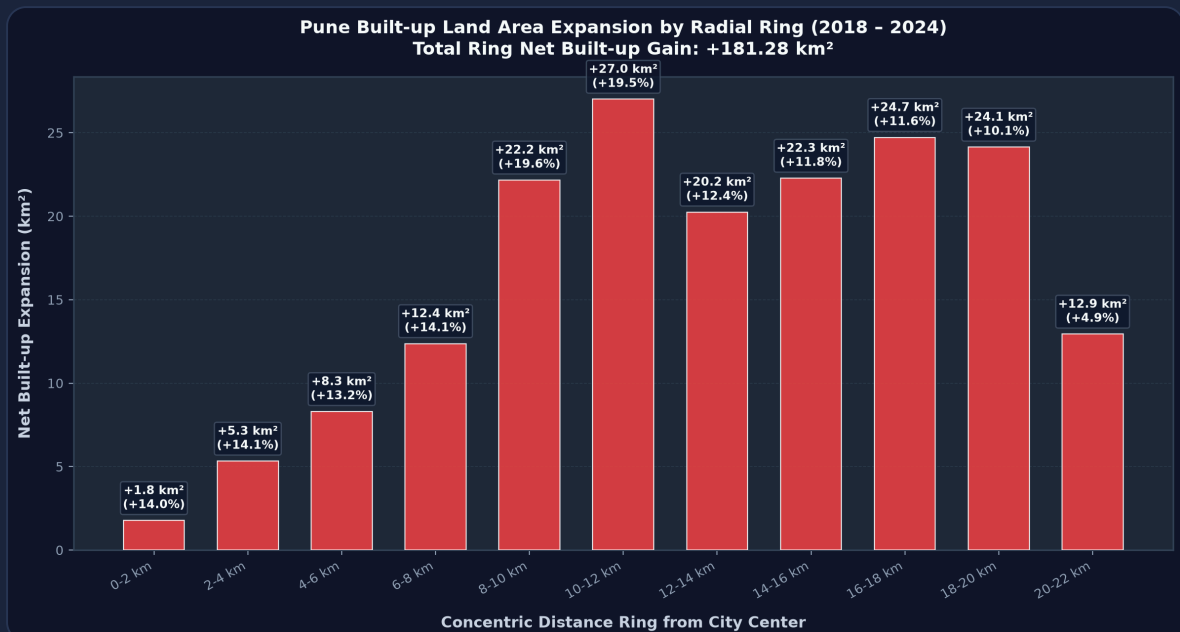
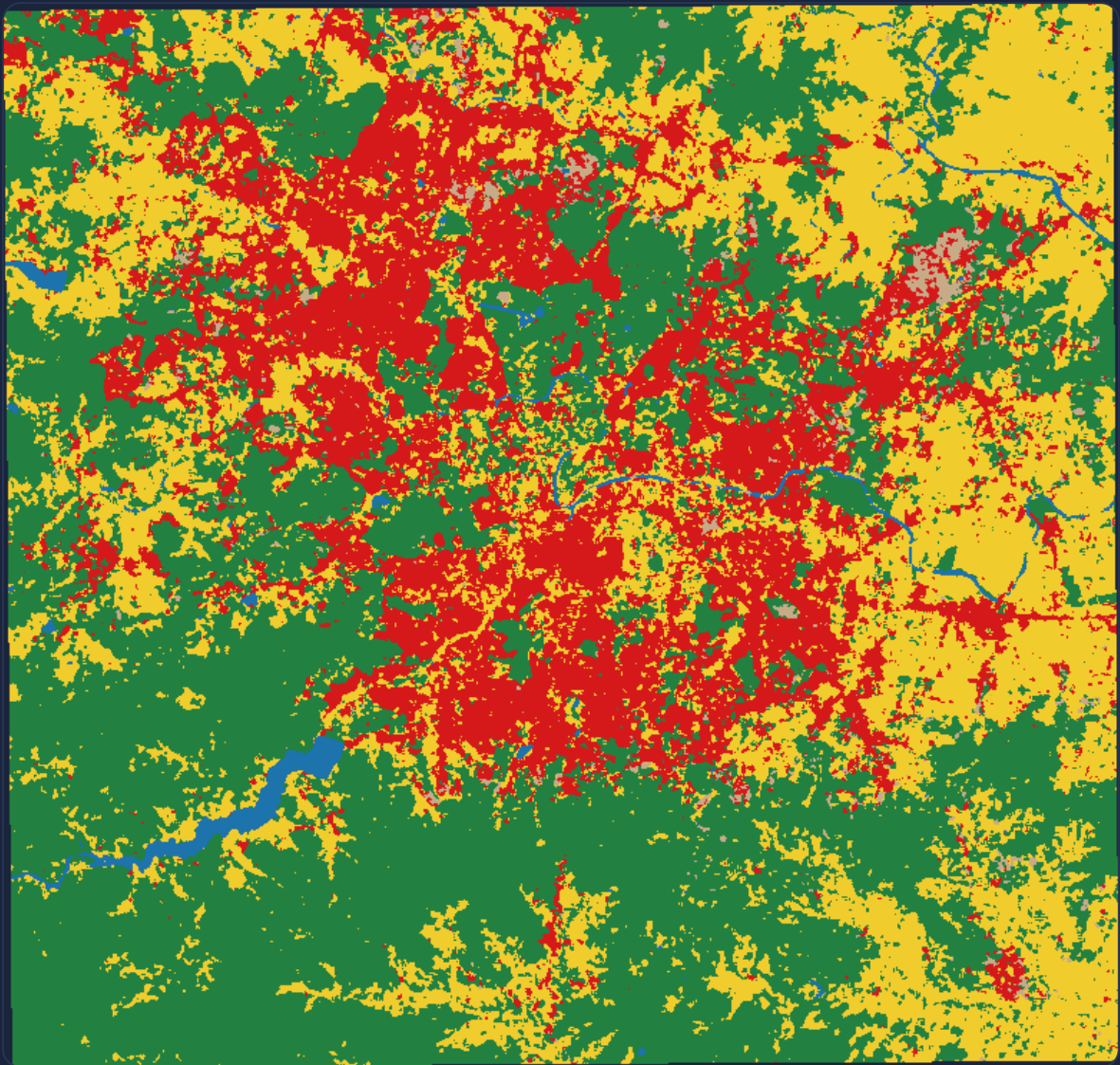


Figure: Pune Annual Ring Growth.

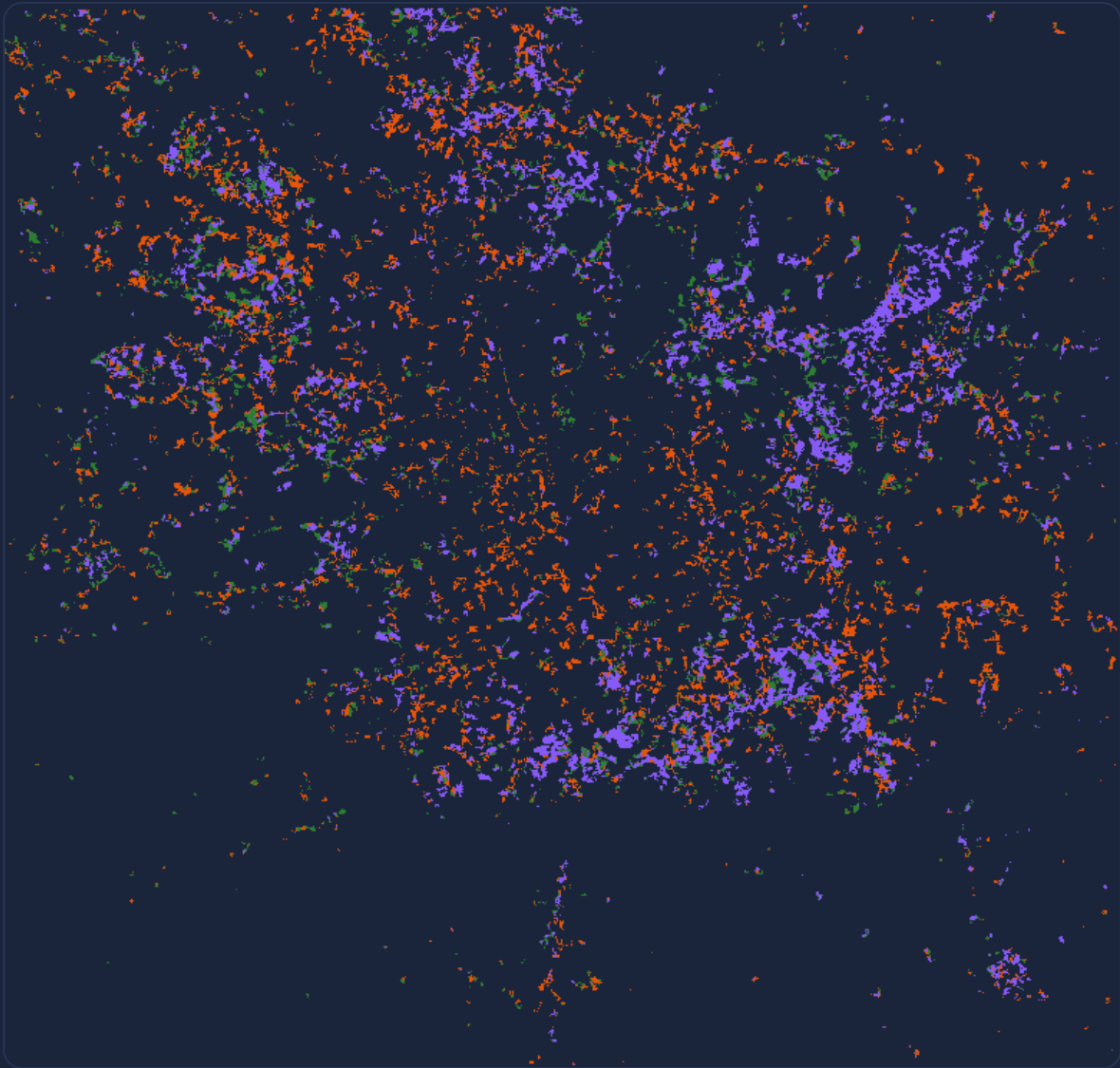
Spatial Sprawl, Entropy & Core-Periphery Distribution

Year	Built-up Area (km ²)	Annual Growth	CAGR vs '18	Shannon Entropy
2018	410.11 km ²	+0.00%	+0.00%	0
2019	326.02 km ²	-20.50%	-20.50%	0
2020	332.13 km ²	+1.87%	-10.01%	0
2021	386.55 km ²	+16.39%	-1.95%	0
2022	692.99 km ²	+79.28%	+14.01%	0
2023	359.65 km ²	-48.10%	-2.59%	0
2024	406.65 km ²	+13.07%	-0.14%	0

Spatial Output Maps



Classified Land Cover (2024)



Land Cover Change (2018 → 2024)



5. Data Quality, Diagnostics & Quality Gate

Every composite is strictly validated for cloud contamination, NoData gaps, radiometric calibration shifts, and year-to-year area volatility before ingestion.

Ahmedabad: Satellite Diagnostics

Year	Scenes	Scene Dates	MGRS Tiles	NoData %	Cloud %	Stable Red
2018	8	2017-12-29, 2018-01-03, 2018-02-12, 2018-03-14	42QZL, 43QBF	0.00%	0.23%	0.1455
2019	8	2018-10-15, 2018-10-25, 2018-11-04, 2019-03-24, 2019-03-29	42QZL, 43QBF	0.00%	0.00%	0.1667
2020	8	2020-01-23, 2020-02-17, 2020-02-22, 2020-02-27, 2020-03-03, 2020-03-08	42QZL, 43QBF	0.00%	0.00%	0.1572
2021	8	2020-11-23, 2020-12-18, 2021-01-12, 2021-02-01, 2021-02-21, 2021-02-26	42QZL, 43QBF	0.00%	0.00%	0.1508
2022	8	2021-10-19, 2021-11-13, 2021-12-18, 2021-12-23	42QZL, 43QBF	0.00%	0.00%	0.1494
2023	8	2022-12-28, 2023-01-22, 2023-01-27, 2023-02-06, 2023-03-03, 2023-03-18	42QZL, 43QBF	0.00%	0.00%	0.1512
2024	8	2023-11-18, 2023-12-23, 2024-01-12, 2024-03-12, 2024-03-22	42QZL, 43QBF	0.00%	0.00%	0.1580

Data Quality Gate Status: **FAILED**

- **Gate 1 (Max NoData $\leq$ 5.0%):** **PASS** — All composites within nominal threshold ($\leq$ 5%)
- **Gate 2 (Max YoY Volatility $\leq$ 15.0%):** **FAIL** — Exceeded in: 2019 (25.7%)
- **Gate 3 (Min Accuracy $\geq$ 70.0%):** **FAIL** — Sub-70% in: 2018 (59.5%), 2021 (68.5%), 2024 (67.8%)
- **Gate 4 (Loss/Gain Ratio $\leq$ 30.0%):** **PASS** — Transition loss verified

Pune: Satellite Diagnostics

Year	Scenes	Scene Dates	MGRS Tiles	NoData %	Cloud %	Stable Red
2018	5	2018-03-06, 2018-03-31, 2018-05-25	43QCA, 43QDA	0.00%	8.28%	0.1452
2019	3	2019-01-05, 2019-01-25	43QCA, 43QDA	0.00%	2.59%	0.1163
2020	8	2019-12-31, 2020-01-30, 2020-02-14, 2020-02-19, 2020-02-24	43QCA, 43QDA	0.00%	0.00%	0.1175
2021	8	2020-11-10, 2020-12-05, 2020-12-25, 2021-01-24, 2021-02-03, 2021-02-28	43QCA, 43QDA	0.00%	0.00%	0.1147
2022	8	2021-11-10, 2021-12-10, 2021-12-20, 2021-12-25	43QCA, 43QDA	0.00%	2.81%	0.1127
2023	8	2022-12-20, 2022-12-25, 2023-01-14, 2023-01-19, 2023-02-03	43QCA, 43QDA	0.00%	0.00%	0.1100
2024	8	2023-11-15, 2024-02-03, 2024-02-18, 2024-02-23, 2024-02-28	43QCA, 43QDA	0.00%	0.00%	0.1225

Data Quality Gate Status: FAILED

- **Gate 1 (Max NoData $\leq$ 5.0%): PASS** — All composites within nominal threshold ($\leq$ 5%)
- **Gate 2 (Max YoY Volatility $\leq$ 15.0%): FAIL** — Exceeded in: 2019 (31.7%), 2021 (18.5%)
- **Gate 3 (Min Accuracy $\geq$ 70.0%): PASS** — All evaluated years $\geq$ 70.0%

- **Gate 4 (Loss/Gain Ratio $\leq$ 30.0%):** **PASS** — Transition loss verified



6. Engineering & Operations

Technology Stack

Layer	Technology	Purpose
EO Ingestion	PySTAC-Client, Stackstac, Dask	Cloud-native STAC scene retrieval & median reduction
ML Pipeline	Scikit-Learn, Rasterio, GeoPandas	Balanced Random Forest classification & spatial filtering
Spatial DB	PostgreSQL 16 + PostGIS 3.4	Spatial envelopes, audit runs, time-series storage
Backend API	FastAPI, Uvicorn, Pydantic	High-throughput REST API with GeoJSON endpoints
Frontend UI	MapLibre GL JS, Vanilla HTML/CSS	WebGL swipe map, time slider, interactive analytics
Orchestration	k3s / k3d, Docker Compose	Lightweight container & Kubernetes cluster parity
CI/CD	GitHub Actions	Automated linting, test execution, Trivy CVE scanning
Monitoring	Prometheus, Grafana	Custom .prom metrics exporter & sprawl monitoring

Test Suite & Payload Metrics

AUTOMATED UNIT & INTEGRATION TESTS

54 Test Cases

100% Passing Coverage across Scaling, Cleanup, Gate & Metrics

UrbanPulse is 100% configuration-driven. Add configs/<city>.yaml with the exact keys:

```
city:
  name: Jaipur
  state: Rajasthan
  country: India
spatial:
  bbox: [75.6000, 26.7000, 76.0500, 27.1500]
  crs: EPSG:4326
  center_lat: 26.9124
  center_lon: 75.7873
  aoi_size_km: 45.0
temporal:
  analysis_years:
    start_year: 2018
    end_year: 2024
  dry_season:
    start_month: 11
    end_month: 2
    start_day: 11-01
    end_day: 02-28
stac:
  earth_search_url: https://earth-search.aws.element84.com/v1
  collections:
    sentinel_2: sentinel-2-c1-l2a
```

7. Limitations

- accuracy is agreement with ESA WorldCover, not field-verified ground truth
- Sentinel-2 covers 2017 onward only
- dry-season composites confuse fallow farmland with built-up land
- the model is not transferable across sensors
- the persistence rule forces non-decreasing built-up area and is a documented assumption
- 10-60 m resolution limits small features.



8. Future Roadmap

1. Historical Landsat Integration (2000–2016)

Extend the temporal horizon by 16 years using harmonized Landsat 5/7/8 surface reflectance stacks with cross-calibration regression.

2. CA-Markov 2035 Urban Growth Forecasting

Couple Cellular Automata and Markov Transition Chains with slope, transport proximity, and zoning constraint layers for 2035 sprawl forecasting.

3. Hand-Labelled Ground-Truth Validation Set

Digitize high-resolution aerial imagery ground truth across 500 stratified random points to measure real-world precision beyond global maps.

4. Dynamic "Run This City" API Endpoint

Expose asynchronous Celery / Redis worker endpoints allowing users to submit any arbitrary bounding box for on-demand sprawl computation.

9. Reproducibility & Verification

All pipelines, tests, and documentation artifacts can be executed locally or in containerized environments with standard commands:

```
# 1. Start full local stack (Web + API + DB + Prometheus)
make up

# 2. Run complete pipeline for a specific city and target year
make pipeline CITY=pune YEAR=2024

# 3. Execute standalone Python pipeline CLI
python -m pipeline.run_city --city ahmedabad

# 4. Run automated test suite
pytest -v

# 5. Re-generate this documentation and PDF report
make docs
```

Component	Version / Status
Python	3.14.2
Rasterio	1.5.1
GeoPandas	1.2.0
Scikit-Learn	1.9.1
Numpy	2.5.3
Pandas	2.3.3
Dask	2026.8.0
Stackstac	0.5.1
PySTAC-Client	0.9.0
Matplotlib	3.11.2
Playwright	Available